

An investigation of Low-cost Air Quality Sensor Calibration to improve their data quality.

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Abstract

The use of low-cost air quality sensors is highly attractive for large-scale deployments due to their affordability, portability, and low power requirements. However, they commonly exhibit limitations related to measurement drift, environmental sensitivity, nonlinear response behaviours, and cross-sensitivity to unrelated gases. To enable reliable use of low-cost sensors in regulatory or scientific applications, calibration against higher-quality reference-grade sensors is essential.

In this paper, we present a novel, extensive calibration analysis, carried out using measurements collected from two sensing nodes: Node 3 (Co-located with TH2 reference station at Queen Mary University London Mile End Road - 51.522530, -0.042155); and Node 5 (Co-located with TH7 reference station at King Edward Memorial Park - 51.509477, -0.050541), each containing low-cost, mid-cost, and high-cost sensing instruments. The two nodes are analysed independently and interactively to evaluate both local calibration (trained and evaluated on the same node) and transfer calibration (trained on one node, applied/tested to another).

The calibration focuses primarily on PM_{2.5}, NO₂, CO, and O₃, as these pollutants have sensor representations across the cost tiers. For each pollutant, the low-cost sensor is calibrated against the corresponding high-cost reference using:

- Linear regression
- Quadratic (second-degree polynomial) regression
- Random Forest regression

We evaluate calibration quality across multiple metrics, including coefficient of determination (R^2), root mean squared error (RMSE), and residual characteristics. Additional analyses include combined-node calibration models and inter-node transferability.

We found that PM_{2.5} was the strongest-performing pollutant across all models and nodes, resulting in a correlation between low-cost and high-cost sensors: 0.70–0.72. This correlation result suggest that low-cost air quality sensors can approximate high-cost PM_{2.5} measurements reasonably well after calibration.

Keywords – Air quality sensing, Linear Regression, Polynomial Regression, Machine learning, Random Forest regression

Introduction

Air quality (AQ) monitoring faces challenges in capturing pollution levels, especially in urban and industrial environments where pollution poses significant threats to health and sustainability. High-grade reference instruments, such as chemiluminescence analysers for NO₂ and TEOM for PM_{2.5}, provide precise, stable data that serve as regulatory baselines. Typically, these reference instruments can achieve accuracies of 1-2% and detect NO₂ concentrations as low as 0.1 ppm. [1] However, these tools are costly (<£10,000), require significant power, and demand intensive maintenance. Due to their expense and maintenance requirements, these instruments are deployed sparsely; for instance, there might be one reference monitoring station every 10 km². [2] This sparse deployment results in networks that often lack sufficient spatial resolution to

characterise hyper-local variability, such as at roadways, schools, or industrial sites. In contrast, low-cost air quality sensors can be deployed much more densely, such as one every 0.2 km², to help fill in these spatial gaps and provide more detailed data.

Low-cost air quality sensors (LCS) have emerged as a promising complementary solution. These sensors—typically based on electrochemical, metal–oxide semiconductor (MOS), or optical particle-counting technologies—are inexpensive (<£500), lightweight, and easy to deploy at high density. However, the data quality they produce is highly variable, often affected by cross-sensitivities, environmental conditions, humidity, temperature, ageing, and differences in individual sensor manufacturing [3] [2]. For these reasons, raw LCS outputs cannot be used directly as substitutes for reference-grade measurements. The central challenge, therefore, is not their availability but their accuracy.

Consequently, calibration—mathematically adjusting the low-cost sensor output to match a trusted reference—is now regarded as the primary mechanism for improving LCS data quality. existing studies highlight several calibration techniques: simple linear regression, multivariate models, polynomial regressions, machine learning models such as Random Forests, and more advanced hybrid or propagation-based methods [3]. When deployed appropriately, calibration can significantly reduce sensor drift, minimise bias, and increase the reliability of low-cost data, bringing it closer to regulatory standards.

Nevertheless, calibration is not always straightforward. First, not all pollutants have reliable reference instruments. Gases such as SO₂, H₂S, and VOCs often lack co-located, continuously running reference-grade devices due to high cost or logistical constraints [2]. This creates gaps in calibration capability and raises an important question: *how can low-cost data for these pollutants be reliably corrected when appropriate reference measurements are unavailable?* Second, calibration is not necessarily a one-time procedure. Low-cost sensors exhibit temporal drift, ageing degradation, and changes in environmental sensitivity over their operational lifetime [3]. Several field studies demonstrate that models calibrated once at the beginning of deployment rapidly degrade after weeks or months unless recalibrated or corrected dynamically. This raises the second core issue: *is calibration a one-off event, or must it be repeated over time to maintain performance?*

The present investigation directly addresses these questions. Using two co-located monitoring nodes (Node 3 and Node 5), each containing low, mid, and high-cost sensors, this study evaluates the extent to which low-cost PM_{2.5}, NO₂, CO, and O₃ measurements can be calibrated against corresponding high-cost reference sensors. The analysis includes linear regression, second-degree polynomial calibration, Random Forest modelling, and cross-node transfer evaluations. Our findings show that polynomial calibration substantially improves alignment for pollutants such as PM_{2.5}, while other species—especially those measured by electrochemical mid-cost sensors (e.g., NO₂ Alphasense A4 series)—exhibit strong sensor-specific behaviours and environmental sensitivity, limiting calibration transferability. Furthermore, results confirm that calibration models trained on one node degrade when applied to another, supporting previous literature that calibration is *sensor-specific and location-dependent*, and therefore must be periodically updated.

Ultimately, this research contributes to an important question for modern air quality science: How can we effectively improve the data quality of low-cost or unknown-quality air quality sensors through calibration against trusted reference instruments—especially when reference sensors are limited, pollutant-specific, or unavailable?

With this paper, we provide a structured investigation of this challenge; evaluate practical calibration strategies; and discuss implications for long-term sensor deployments, algorithmic correction, and the future reliability of dense low-cost air quality monitoring networks.

Cost-Based Classification Framework

According to Giordano et al. (2021) [1], air quality sensors can be broadly categorised by their cost, which tends to reflect differences in performance, reliability, and intended application:

- **Low-Cost Sensors (< \$500 USD):** This group includes sensors like the Sensirion SPS30, Plantower PMS, and various MQ-series metal-oxide or basic electrochemical sensors. These sensors are popular due to their affordability and ease of use, which makes them ideal for widespread deployment in dense networks. However, they are prone to environmental influences (e.g., temperature, humidity) and show high variability between individual units, which can compromise data accuracy.
- **Mid-Range Sensors (\$500–\$10,000 USD):** Represented mainly by advanced electrochemical models such as the Alphasense B4 series, this category offers better performance in terms of stability, sensitivity, and reduced drift. Despite these improvements, achieving accurate readings often requires more sophisticated calibration methods that account for multiple environmental variables.
- **High-End Reference Instruments (> \$10,000 USD):** Used in regulatory and governmental air quality monitoring, these instruments (e.g., beta-attenuation monitors, chemiluminescence analysers) provide the highest levels of precision, accuracy, and traceability. However, their high cost, size, and maintenance requirements limit their use to professional and institutional settings.

Related Work

Low-cost air quality (AQ) sensors expand environmental monitoring, especially in areas lacking costly reference equipment. Yet, measurement inaccuracy, environmental interference, and calibration needs, limit their broader use. Without proper calibration, these sensors often display bias, drift, and site-dependent performance, as seen across particulate matter (PM), nitrogen dioxide (NO₂), ozone (O₃), and carbon monoxide (CO). This section reviews calibration methods for low-cost sensors targeting these pollutants.

Challenges in Low-Cost AQ Sensor Accuracy

Environmental and Operational Factors

Giordano et al. (2021) [1] highlight that PMS-based optical particle sensors are particularly sensitive to environmental factors, including relative humidity, aerosol composition, and particle refractive indices. These factors complicate the conversion of particle count to mass concentration, often leading to systematic over- or underestimation.

Cross-sensitivity poses a significant challenge for electrochemical NO₂, O₃, and CO sensors. For example, Zuidema et al. (2021) [2] found that NO₂ sensors respond strongly to O₃, whereas CO sensors are influenced by humidity and temperature changes. As a result, raw readings require correction using compensation models, such as linear or multivariate regression.

Sensor-to-Sensor Variability and Drift

Sensor drift is consistently reported across studies.

- Huang et al. (2022) [4] observed performance degradation in low-cost PM and NO₂ sensors during field deployments, requiring periodic recalibration.
- Similarly, Rogulski et al. (2022) [3] reported that Alphasense NO₂ sensors show complex, non-linear drift over time, which can be partially addressed using mathematical models.

Calibration for low-cost sensors must be ongoing, with regular recalibration throughout the operational life to maintain accuracy.

Calibration Approaches in the Literature

Laboratory vs. Field Calibration

Giordano et al. (2021) [1] note that laboratory calibrations often do not translate to real-world conditions due to environmental differences. Consequently, calibrating sensors alongside regulatory-grade monitors in the field is now regarded as the most reliable approach.

Zuidema et al. (2021) [2] conducted an extensive field deployment in which electrochemical CO, NO, NO₂, and O₃ sensors were installed next to reference analysers for several months. Their findings underscore the importance of cross-validation at different sites to produce calibration models with broader applicability.

Statistical and Machine Learning Methods

Across studies, three broad calibration strategies appear:

Linear and Polynomial Regression

Used widely due to transparency and interpretability.

- Huang et al. (2022) [4] applied multivariate regression for PM sensors using humidity and temperature corrections.
- Rogulski et al. (2022) [3] successfully used polynomial regression to correct Alphasense NO₂ outputs, showing significant improvements in R² and reduced RMSE after calibration.

These studies support the use of polynomial regression as an effective calibration method for gas sensors with non-linear responses, a focus of this dissertation.

Advanced Machine Learning Techniques

Vajs et al. (2023) [5] introduced a machine learning-based system called “calibration propagation.” This technique allows sensor networks to maintain calibration even when some sites are not collocated with reference instruments. Their method integrates:

- Random forests
- Neural networks
- Transfer learning across sensor nodes.

This demonstrates the trend toward adaptable, data-driven calibration models that reduce dependence on costly reference infrastructure.

Calibration Transfer and Network-Level Reliability

The concept of **calibration transfer**—training a model on one sensor and deploying it across other sensors or nodes—is increasingly studied.

- Zuidema et al. (2021) [2] found that the effectiveness of a calibration model when deployed at new locations depends on both the pollutant type and the degree of similarity in environmental conditions across sites.
- To address portability, Vajs et al. (2023) [5] suggest using machine learning propagation as a potential solution for transferring calibration between sensor sites.

A central focus of this dissertation is whether *calibration models trained on one sensor node can maintain accuracy when applied to other nodes.*

Calibration Limitations and Practical Considerations

Availability of Reference Standards

Some pollutants, such as SO₂ and certain volatile organic compounds (VOCs), lack readily deployable reference analysers. This limits calibration opportunities and reinforces the need for transferable, network-level-calibrated approaches (Giordano et al., 2021) [1].

Frequency of Calibration

The literature is clear:

- Calibration is an ongoing requirement, not a single event.
- Factors such as sensor ageing, environmental deposition, component degradation, and seasonal changes all make regular recalibration necessary (Huang et al., 2022; Rogulski et al., 2022). [4] [3]
- Although machine learning propagation (Vajs et al., 2023) [5] can help reduce the need for frequent recalibration, it cannot eliminate the need for at least some reference measurements.

The literature establishes that low-cost sensors offer unprecedented spatial coverage but inherently lack the stability, precision, and linearity of reference-grade instruments. Calibration strategies—especially polynomial regression, random forest models, and ML calibration propagation—significantly improve their performance. However, challenges persist due to environmental variability, sensor drift, and limited reference infrastructure.

These findings lay the foundation for the present study, which investigates whether polynomial calibration, linear regression, and more advanced modelling approaches can reliably improve the accuracy of low-cost sensors, whether models trained on one node transfer successfully to another, and how calibration equations differ across sensor types and deployment environments.

Research Gaps and MSc Project Contribution

This comprehensive literature review reveals several key research gaps that are directly aligned with the objectives of the proposed MSc project.

- **Lack of Systematic Inter-Comparison Across All Three Cost Tiers:** The literature contains many studies comparing low-cost to high-cost sensors, and some comparing mid-cost to high-cost. However, there is a scarcity of research that systematically deploys and evaluates low, mid, and high-cost sensors simultaneously under identical environmental conditions. This project's core design directly addresses this gap.
- **Validating and Extending Calibration Propagation in Hybrid Networks:** The concept of multi-hop calibration propagation (Vajs et al., 2023) [5] is a promising but nascent field. Key questions remain: How does accuracy degrade in a more complex, hybrid network? Can mid-cost sensors act as stable 'calibration hubs' to anchor the calibration of surrounding low-cost sensors? This project is perfectly positioned to test these hypotheses.
- **Development of Integrated and Transferable Calibration Models:** A significant opportunity exists to develop integrated calibration models that leverage the entire network's data (e.g., using data from a mid-cost sensor to improve a low-cost sensor's model). Furthermore, the transferability of models between different sites and seasons remains a major challenge that requires investigation into transfer learning or meta-learning approaches.
- **Long-Term Performance and Adaptive Calibration:** Many studies are limited to short-term deployments. Characterizing sensor drift and seasonal performance variations over periods greater

than one year is crucial. This enables the development of adaptive models that can automatically detect and correct for drift without manual intervention.

- **Uncertainty Quantification:** A systematic characterization of how uncertainty propagates from individual sensor measurements to final, network-level air quality estimates is largely missing from the literature. This project can quantify the trade-offs between network density, sensor cost, and overall uncertainty.

Methodology

Motivated by prior findings that low-cost sensors exhibit bias, drift, and environmental sensitivity (Giordano et al., 2021; Huang et al., 2022; Zuidema et al., 2021) [1] [4] [2], we evaluate multiple calibration strategies using real-world sensor deployments. The approach combines data preprocessing, temporal alignment, model development, validation, and cross-node transfer testing to assess the accuracy and generalisability of calibration models.

Requirements

Two sensor nodes—Node 3 and Node 5—serve as the case studies for this investigation. The full dataset was obtained from the publicly accessible GitHub repository provided by *Andrés Aharhel Mercado Velázquez* [6]. The repository was downloaded into a secure Google Drive environment to facilitate cloud-based analysis. All data processing, model development, and calibration experiments were conducted within a dedicated Google Colab notebook, enabling version-controlled computational workflows.

Data

The dataset originates from a multi-node air quality monitoring deployment consisting of high-, mid-, and low-cost sensors grouped by pollutant type. Each node contains measurements in *csv* files for PM2.5, NO₂, CO, O₃, and environmental covariates such as temperature and humidity.

Data Pre-Processing

We normalised all *csv* data by combining the date and time fields into a single index called *datetime*. We then removed all missing timestamps (Not a Number, NaNs) and aligned low, mid and high-cost sensors on a time basis.

Nearest-timestamp alignment was applied to both nodes to ensure that each reference point had the closest matching low-cost value. This alignment method is commonly used in field deployments. [4] [2]

We called *load_node_data* function to perform this.

To evaluate calibration model performance, we partitioned each node dataset into 20% Training, 20% Validation and 60% Testing, using a randomised split.

We performed various correlation plots and matrices to determine the best performing pollutant in relation to their mid, low, high – cost value.

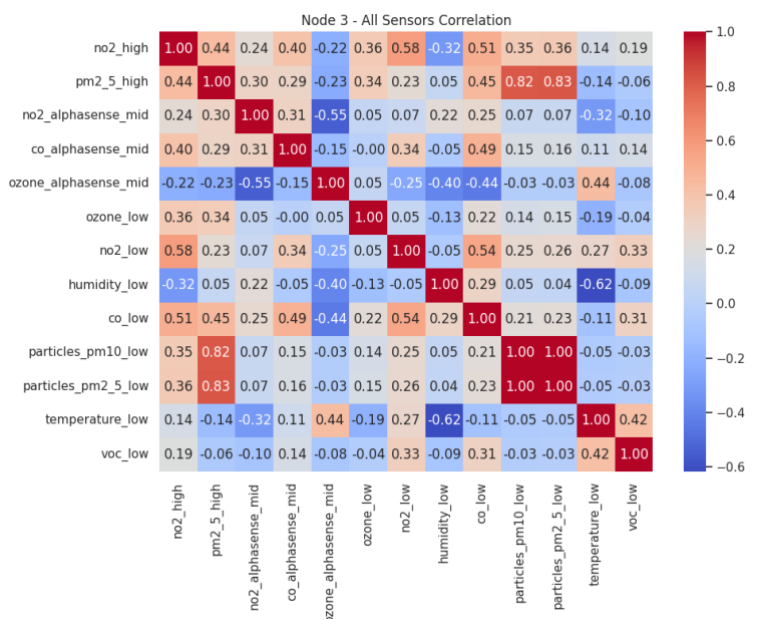


Figure 1: Node 3 - All Sensors Correlation. PM2.5 (High vs Low) — STRONG correlation ($r \approx 0.83$). Low-cost PM2.5 correlates very well with the high-cost reference meaning the low-cost PM2.5 sensors are reliable and can be calibrated using high-cost data. The best.

A. Linear Regression

Pollutant	High vs Low	Mid vs Low	Notes
PM2.5	0.83	moderately good	Best pollutant
NO ₂	0.58	0.07	Mid sensor broken
CO	–	~0.49	reasonable
Ozone	–	~0.05	unusable
Humidity/Temp/VOC	good	good	sensors consistent

Table 1: Pollutants correlations analysis

as a first calibration step for low-cost PM and gas sensors. [4] Findings are visualised in Figure 2, Figure 3 and Figure 4 in the Appendix.

B. Second Degree Polynomial Regression

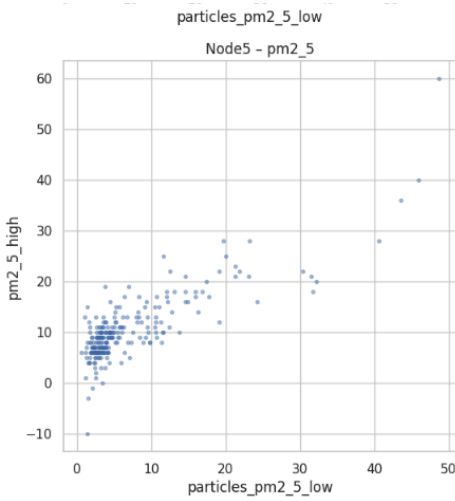


Figure 5: Node 3 second degree polynomial regression model implemented on PM2.5.

Given that many low-cost sensors behave non-linearly—especially electrochemical NO₂ and CO sensors due to cross-sensitivities [3], a quadratic polynomial model was employed: $y_{ref} = ax_{low}^2 + bx_{low} + c$ where y_{ref} in this case, is the pollutant’s targeted high value, and x is the value of the low-cost reference data.

This model produced coefficients such as: $PM2.5_{high} = 0.01147x^2 + 1.0433x + 2.0098$ for Node 3 PM2.5 calibration. $x = PM2.5_{low}$

Due to the negative quadratic term meaning that the curve bends down at high values. This suggests the Node 3 low-cost PM2.5 sensor over-responds at high concentrations.

C. Random Forest Regression

Random Forest (RF) was configured with 200 decision trees, minimum samples per leaf = 5, and depth selected via a validation set. It was used to capture non-linear pollutant behaviour, sensor drift and environmental dependencies. This aligns with recent research where RF and other ML models outperform simple regressions, particularly under variable real-world conditions [5] [2].

Evaluation

In order to have a thorough evaluation of our model, we performed three modelling strategies for each pollutant:

1. Node-specific calibration model (we train, validate and test on the same Node)
2. Cross-node transfer calibration model (Node 5 model applied to Node 3, and vice versa).
3. Combined calibration model (we train on combined Node 3 + Node 5).

We evaluated PM2.5 in both Nodes, and below are the performance tables:

Dataset	R ²	RMSE
Train	0.574	3.02

Validation	0.097	3.86
Test	0.644	4.889

Table 2 Performance Node 3 (Node-specific calibration)

The model performs reasonably well on the test set, indicating the $PM_{2.5}$ relationship is approximately quadratic. Low validation accuracy suggests potential temporal drift or non-stationarity inside the node. Good test performance indicates the learned calibration shape is generalisable.

For Node 5:

Dataset	R^2	RMSE
Train	0.521	3.26
Validation	0.503	3.67
Test	0.720	4.038

Table 3 Performance Node 5 (Node-specific calibration)

Node 5 produces the most stable and consistent calibration results across all splits. The high test R^2 (0.720) indicates that Node 5 low-cost $PM_{2.5}$ is highly predictable. The positive quadratic term suggests slightly different low-cost sensor behaviour compared to Node 3.

Testing how well a model from one node performs on another allows evaluation of sensor reproducibility and cross-deployment generalisation. With the Cross-Node transfer model, Node 5's model performs poorly on Node 3, indicating: Node 5 low-cost behaviour is more stable; Node 3 low-cost sensor shows greater variance and potentially more drift.

Cross-node transfer degradation was expected since environmental conditions and sensor ageing differ between nodes.

The combined model performs exceptionally well on Node 5, even better than the Node-5-only model; On Node 3, performance decreases slightly ($R^2=0.485$), indicating Node 3 is harder to calibrate.

Performance

$PM_{2.5}$ low-cost sensors can be effectively calibrated using simple polynomial models. Achieved test R^2 values: 0.64 – 0.73, which is strong for low-cost sensor correction.

Node 5 sensors show more consistent behaviour, both internally and when used for combined modelling. Node 3 has more noise and drift, resulting in lower validation and cross-node performance.

Calibration curves are not identical across nodes, proving that each low-cost sensor has unique response characteristics, and that the ambient conditions and installation differences affect readings.

Conclusion and Further Work

Calibration can substantially improve low-cost air quality sensor data, especially for particulate matter. However, gas sensors— NO_2 , CO, and O_3 —require more careful, frequent, and node-specific recalibration. A universal calibration approach is viable only for $PM_{2.5}$. Ultimately, integrating reference sensors with machine-learning calibration propagation offers a scalable path toward high-quality air pollution monitoring networks.

With this paper, we present an approach to calibrate $PM_{2.5}$, providing a high R^2 (0.6 -0.75).

Future work will be to incorporate temperature and humidity in the data and explore neural-network or Gaussian calibration models. Finally, our dataset only provided two nodes: expanding to multi-node requires further investigation.

Appendix

Summary Table of Key Papers

The following table summarizes the five most impactful papers from the literature review, selected for their direct relevance to the PhD project's objectives.

<i>APA Reference (7th ed.)</i>	<i>Sensors Compared</i>	<i>Calibration/ Methods</i>	<i>Key Findings</i>	<i>Relevance to MSc Project</i>
<i>Giordano, M. R., et al. (2021).</i>	Primarily PM sensors (various low-cost tiers) vs. reference monitors.	Comprehensive review of methods from linear regression to machine learning.	Provides a foundational framework. Emphasizes the critical impact of Relative Humidity (RH) and the need for robust, site-specific calibration and cross-validation.	High. Offers a guide to best practices and highlights the core challenges the project must address, particularly regarding PM sensor calibration.
<i>Zuidema, C., et al. (2021). ..</i>	Alphasense (CO, NO, NO ₂ , O ₃) vs. regulatory reference stations.	Multiple Linear Regression (MLR) with sensor-specific intercepts, T, RH, and co-pollutant terms.	Argues for MLR over complex ML for interpretability and robustness. Achieved very high performance (R^2 up to 0.97) by including sensor-specific IDs and co-pollutants.	High. Provides a robust, validated, and alternative methodology for calibrating the project's mid-cost Alphasense sensors, challenging the 'ML is always better' assumption.
<i>Vajs, I., et al. (2023).</i>	Alphasense NO2-B43F, Plantower PMS7003 vs. reference station.	Random Forest (RF) used for a novel multi-hop calibration propagation method.	Demonstrates that calibration can be 'propagated' through a network. The method is highly robust for PM sensors but degrades for NO ₂ after 3-4 'hops'.	Critical. Introduces a state-of-the-art, logistically efficient calibration strategy that the project can adopt and investigate further for its own hybrid network.
<i>Agrawal, A., et al. (2024)</i>	Sensirion SPS30, Plantower PMS5003, and others vs. reference instruments.	Linear Regression, Polynomial Regression, Artificial Neural Network (ANN).	The Sensirion SPS30 showed the best performance and consistency. ANN provided the most accurate generic calibration model ($R^2 > 0.91$).	High. Directly validates the choice of the SPS30 as a high-quality low-cost sensor and confirms the effectiveness of ANN for its calibration.
<i>Rogulski, M., et al. (2022).</i>	Alphasense NO2-B43F vs. reference stations.	Custom mathematical correction function incorporating raw sensor signals, T, and RH.	Identifies a critical failure of manufacturer formulas, causing negative readings at high temperatures. The proposed model eliminates this issue and significantly improves accuracy.	Critical. Reveals a fundamental flaw in mid-cost sensors that the project must test for and address, providing a clear methodology to do so.

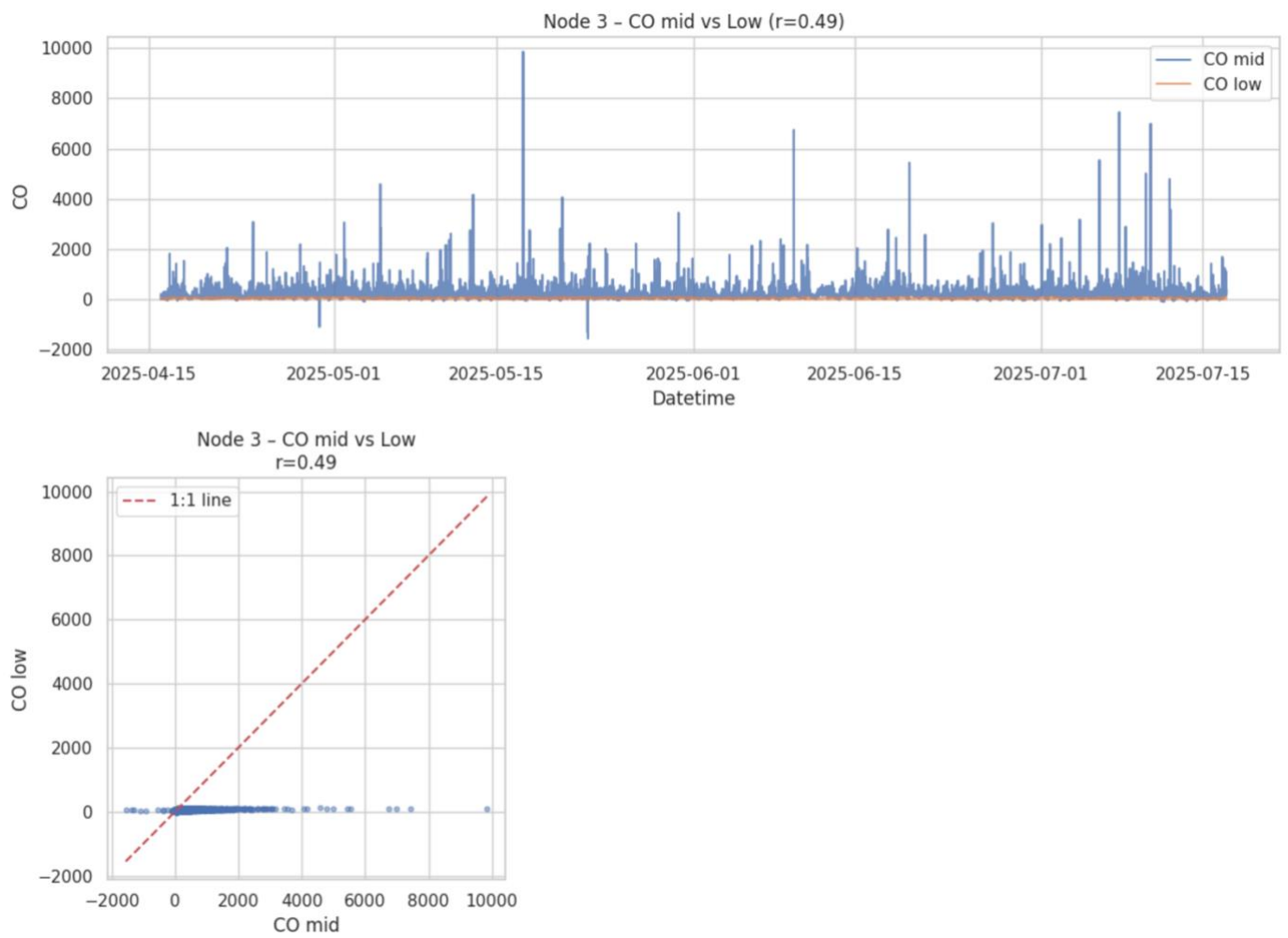


Figure 2: Node 3 - CO mid vs Low analysis

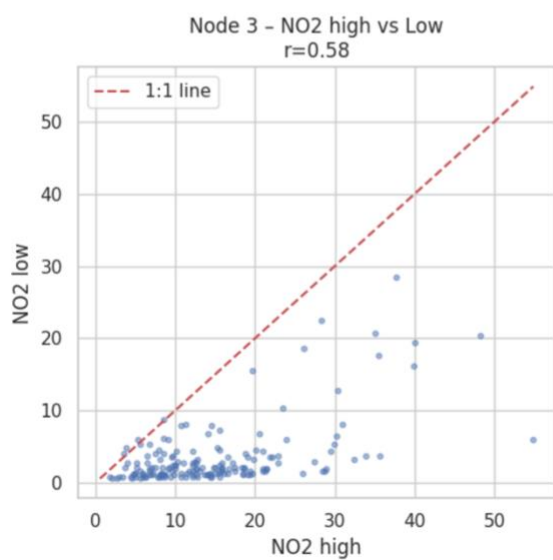
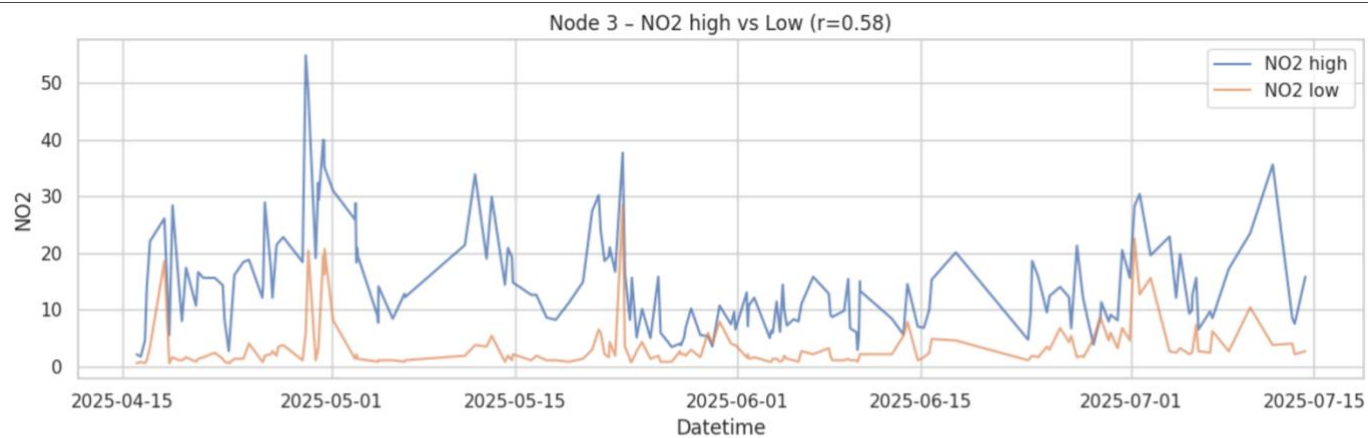


Figure3 3:Node 3 - NO2 high vs Low analysis

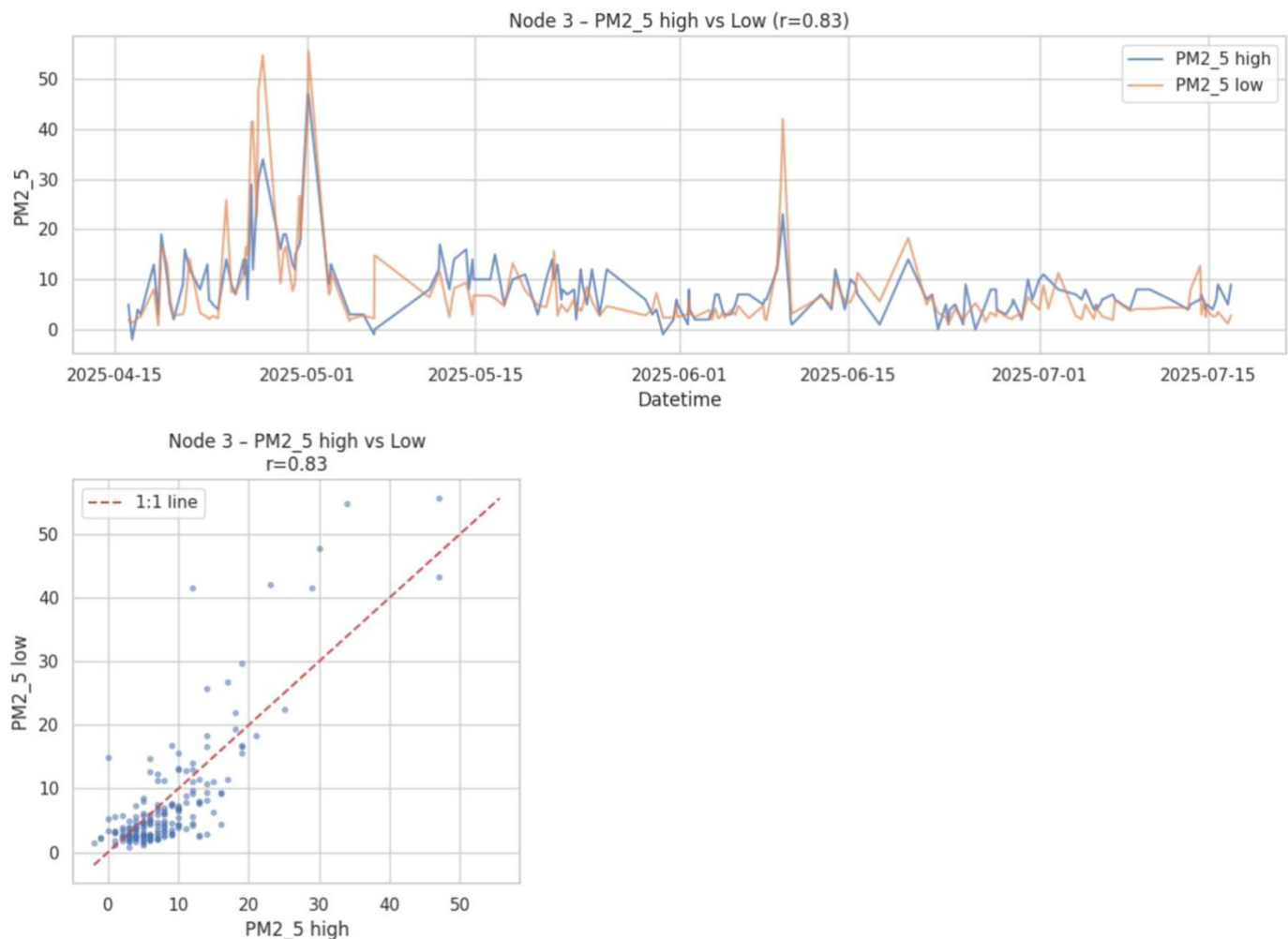


Figure4 4: PM2.5 high vs low analysis

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